

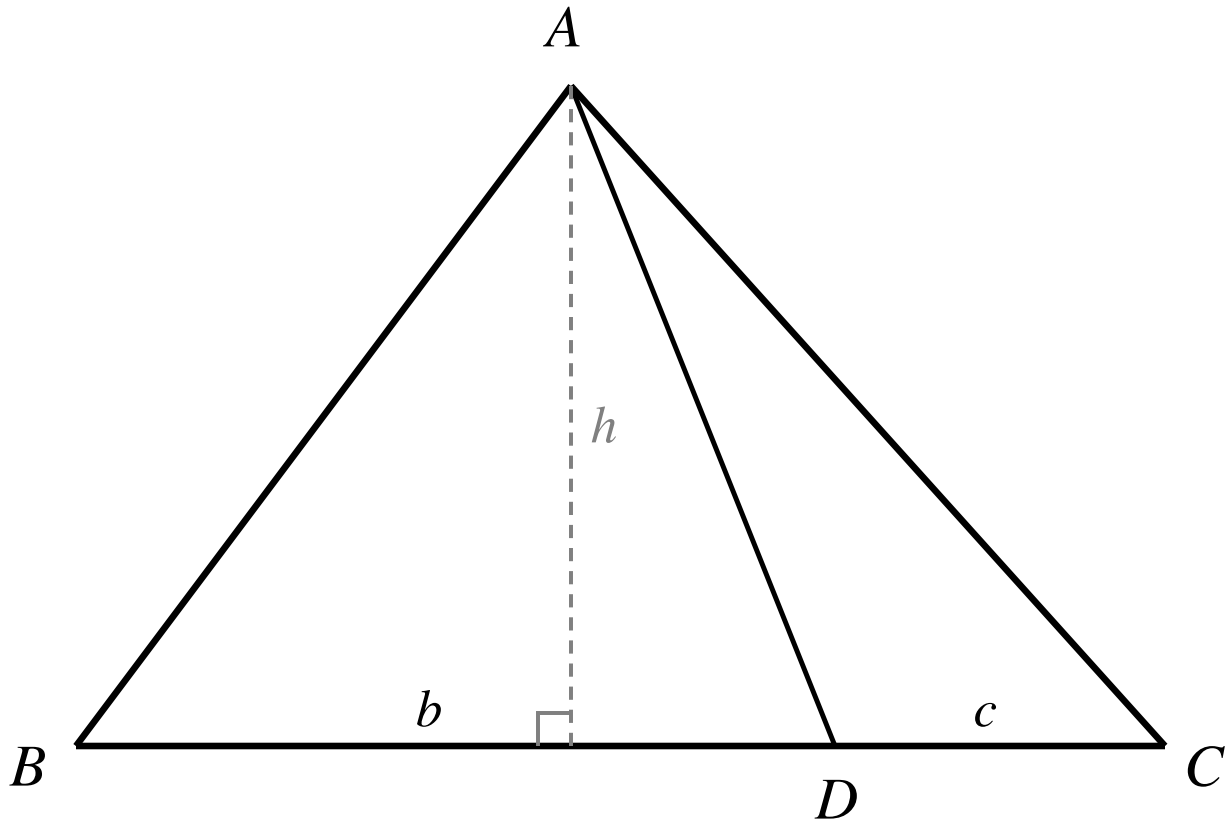
Lecture 13: Area and Triangles

Area of a Triangle: The area of a triangle is the half of the parallelogram formed by doubling the triangle along one of its sides. If we take a triangle with base b and height h , the area is given by:

$$A = \frac{1}{2}bh$$

Lemma 1: Let ABC be a triangle, and let D be a point on the line BC . Then we have the area ratio formula:

$$\frac{S_{\triangle ABD}}{S_{\triangle ADC}} = \frac{BD}{DC}$$



Proof:

- Let h be the height of $\triangle ABC$ with respect to the base BC , $b = BD$ and $c = DC$.
- Then we have $S_{\triangle ABD} = \frac{1}{2}bh$ and $S_{\triangle ADC} = \frac{1}{2}ch$. Thus, we have

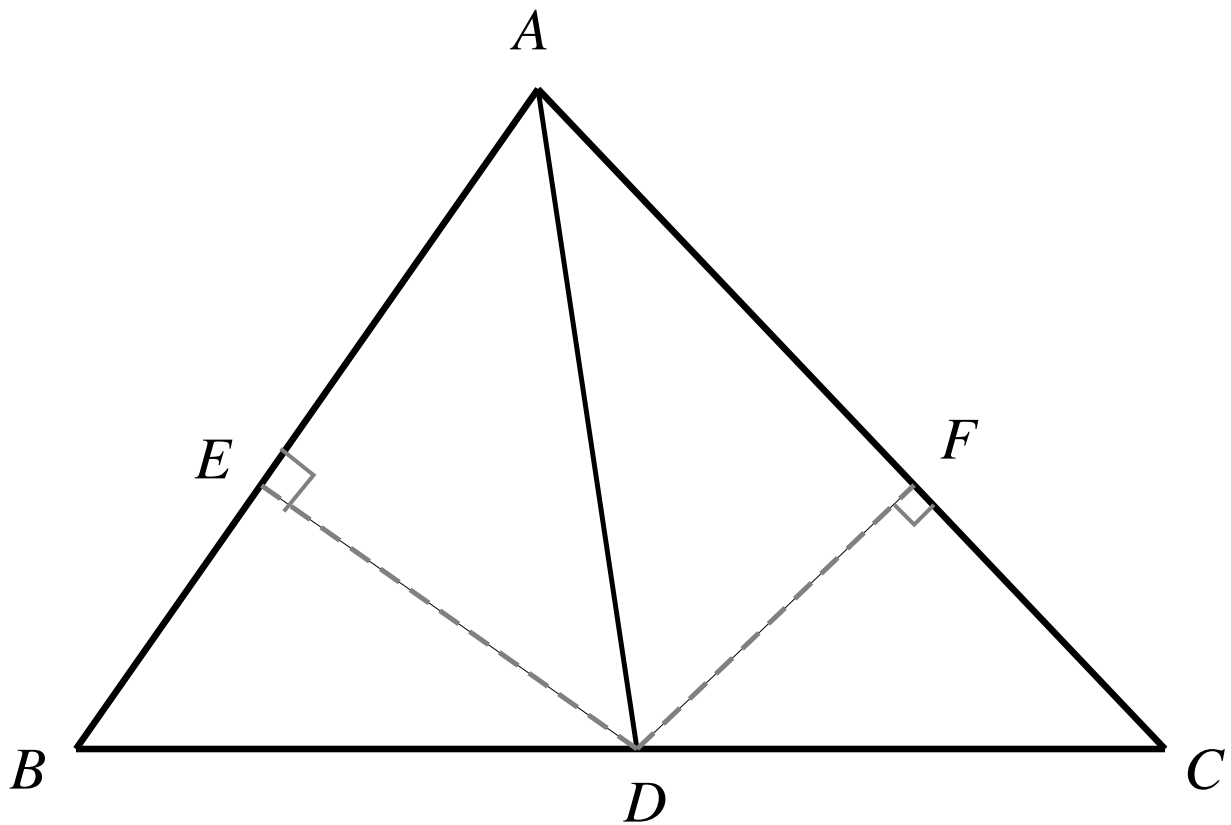
$$\frac{S_{\triangle ABD}}{S_{\triangle ADC}} = \frac{\frac{1}{2}bh}{\frac{1}{2}ch} = \frac{BD}{DC}.$$



Theorem 2 (Bisector theorem) : Let ABC be a triangle, and let AD be the bisector of $\angle BAC$. Then we have the length ratio formula:

$$\frac{BD}{DC} = \frac{AB}{AC}$$

Conversely, if D is a point on the line BC such that $\frac{BD}{DC} = \frac{AB}{AC}$, then AD is the bisector of $\angle BAC$.



Proof:

- Draw two altitudes from D to the lines AB and AC , and let the feet of the altitudes be E and F , respectively.
- First we show that $\triangle AED$ is congruent to $\triangle AFD$. Indeed, by **ASA**, we have $\angle EAD = \angle FAD$, $\angle ADE = \angle ADF$ and AD shared. Thus, $\triangle AED \cong \triangle AFD$. From the congruence, we get $DE = DF$.

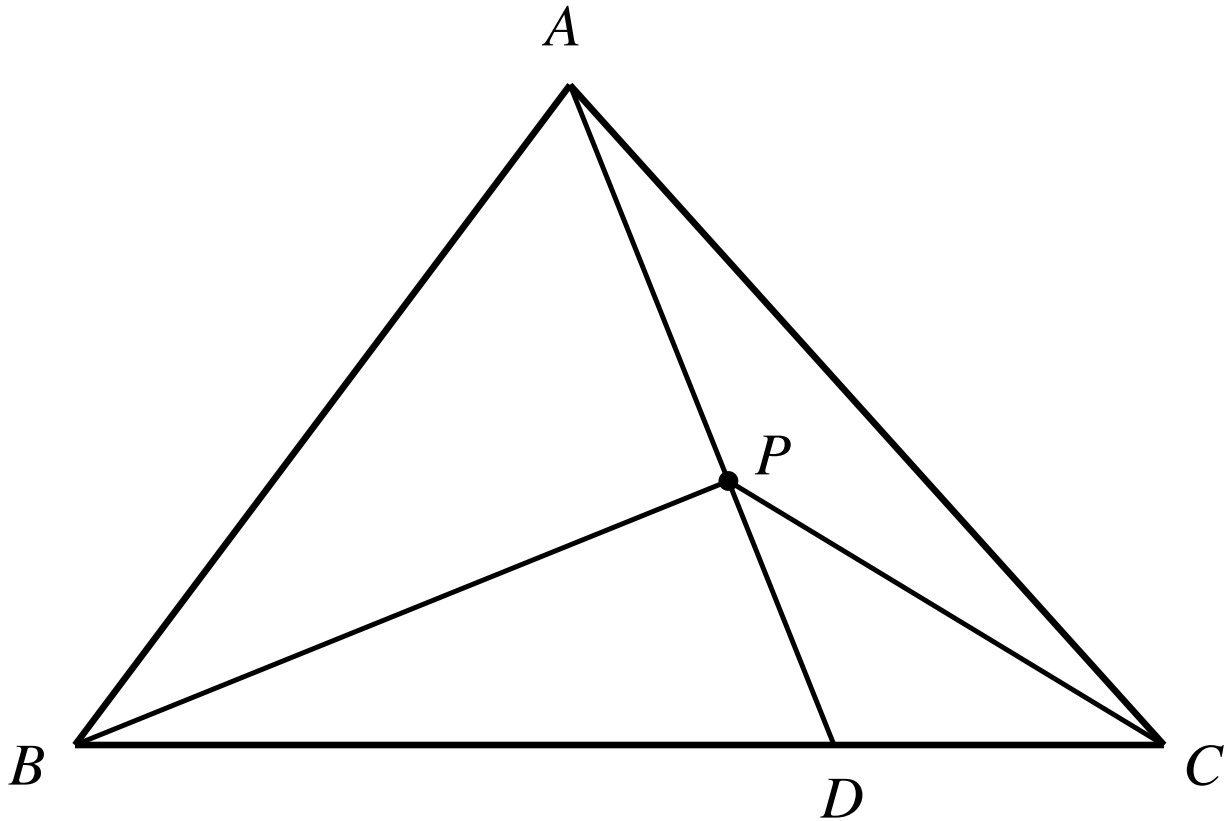
- Now we have

$$\frac{BD}{DC} = \frac{S_{\triangle ABD}}{S_{\triangle ADC}} = \frac{\frac{1}{2}DE \cdot AB}{\frac{1}{2}DF \cdot AC} = \frac{AB}{AC}$$

- Conversely, if $\frac{BD}{DC} = \frac{AB}{AC}$, then we have $DE = DF$ by the same area ratio formula. Now $AE = AF$ by the Pythagorean theorem, and AD is shared. Thus, $\triangle AED \cong \triangle AFD$ by **SSS**, which implies that $\angle EAD = \angle FAD$. Thus, AD is the bisector of $\angle BAC$. ■

Lemma 3 (Seva's Lemma): With the same setup as in Lemma 1, and let P be a point on the line AD . Then we have the area ratio formula:

$$\frac{S_{\triangle ABP}}{S_{\triangle ACP}} = \frac{BD}{DC}$$



Proof:

- By Lemma 1, we have

$$\frac{S_{\triangle BDP}}{S_{\triangle DCP}} = \frac{S_{\triangle ABD}}{S_{\triangle ADC}} = \frac{BD}{DC}$$

- So the difference of the two triangles also satisfies the same ratio, i.e.

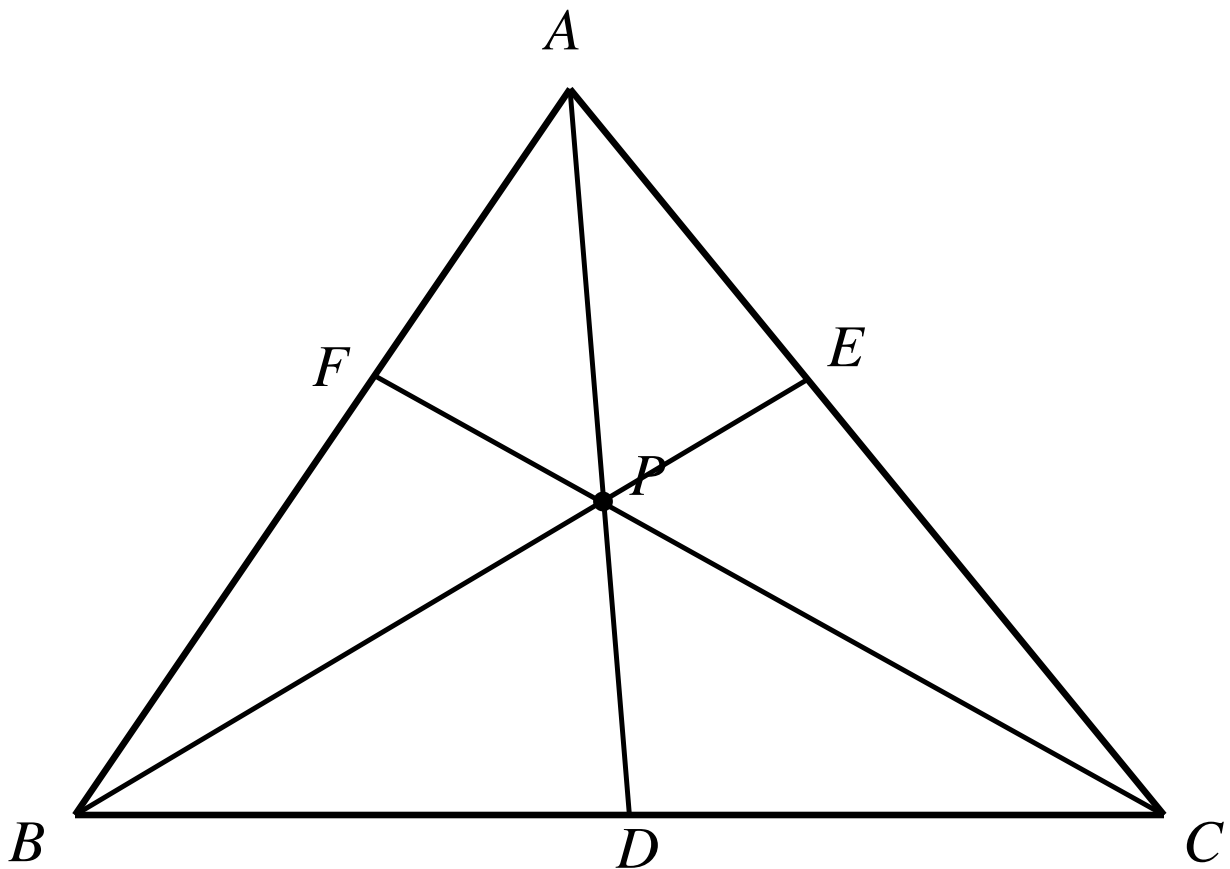
$$\frac{S_{\triangle ABP}}{S_{\triangle ACP}} = \frac{BD}{DC}$$



Theorem 4 (Seva's Theorem): Let ABC be a triangle, and let P be a point in the interior of $\triangle ABC$. Let D , E and F be the intersections from AP , BP and CP to the lines BC , CA and AB , respectively. Then we have the following product formula:

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1$$

Conversely, if D , E and F are points on the lines BC , CA and AB such that the above product formula holds, then the lines AD , BE and CF are concurrent.



Proof:

- By Lemma 3, we have

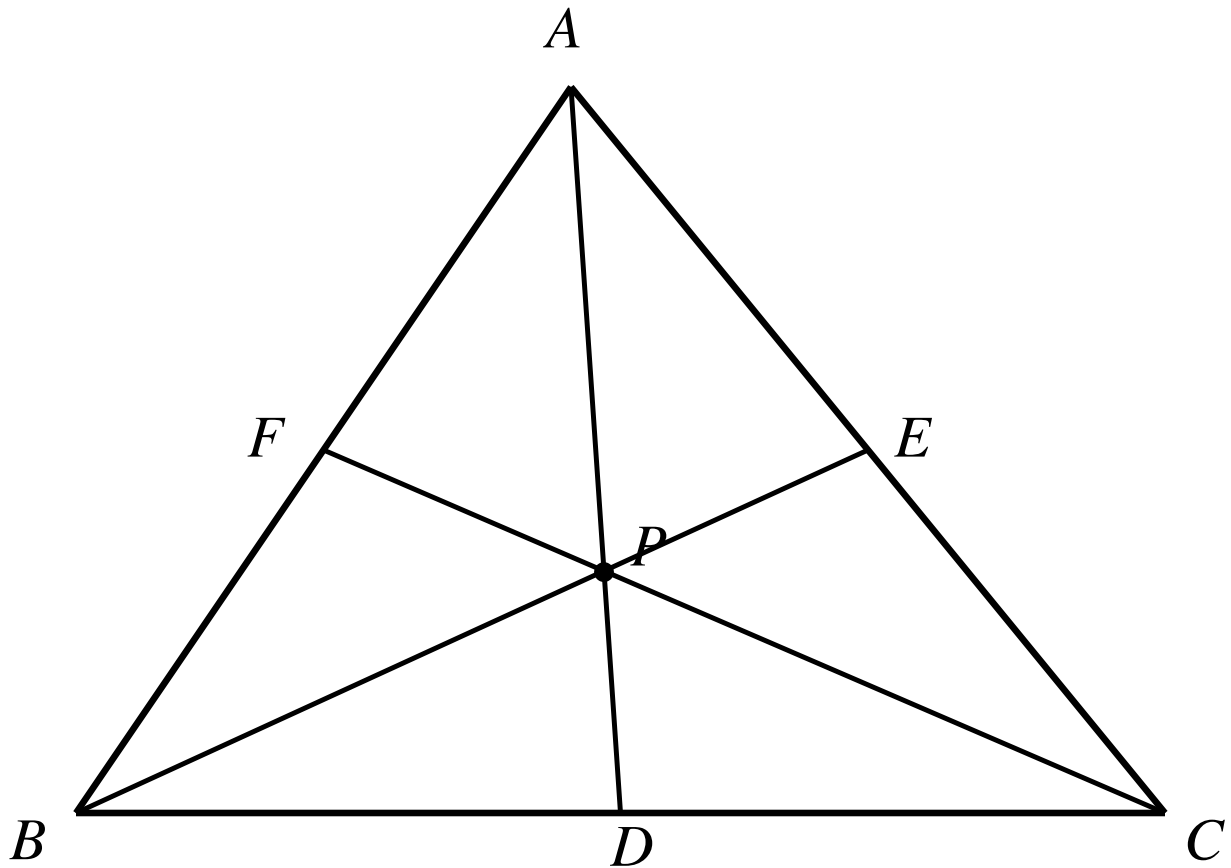
$$\frac{S_{\triangle ABP}}{S_{\triangle ACP}} = \frac{BD}{DC}, \quad \frac{S_{\triangle BCP}}{S_{\triangle ABP}} = \frac{CE}{EA}, \quad \frac{S_{\triangle CAP}}{S_{\triangle BCP}} = \frac{AF}{FB}$$

- Multiplying the three equations together, we get

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = \frac{S_{\triangle ABP}}{S_{\triangle ACP}} \cdot \frac{S_{\triangle BCP}}{S_{\triangle ABP}} \cdot \frac{S_{\triangle CAP}}{S_{\triangle BCP}} = 1$$

- Conversely, if the product formula holds, then we can construct P as intersection of AD and BE , then CP intersects AB at F' cutting AB at the same ratio. So $F = F'$. ■

Theorem 5: Three medians of a triangle intersect at a single point (the **centroid**).



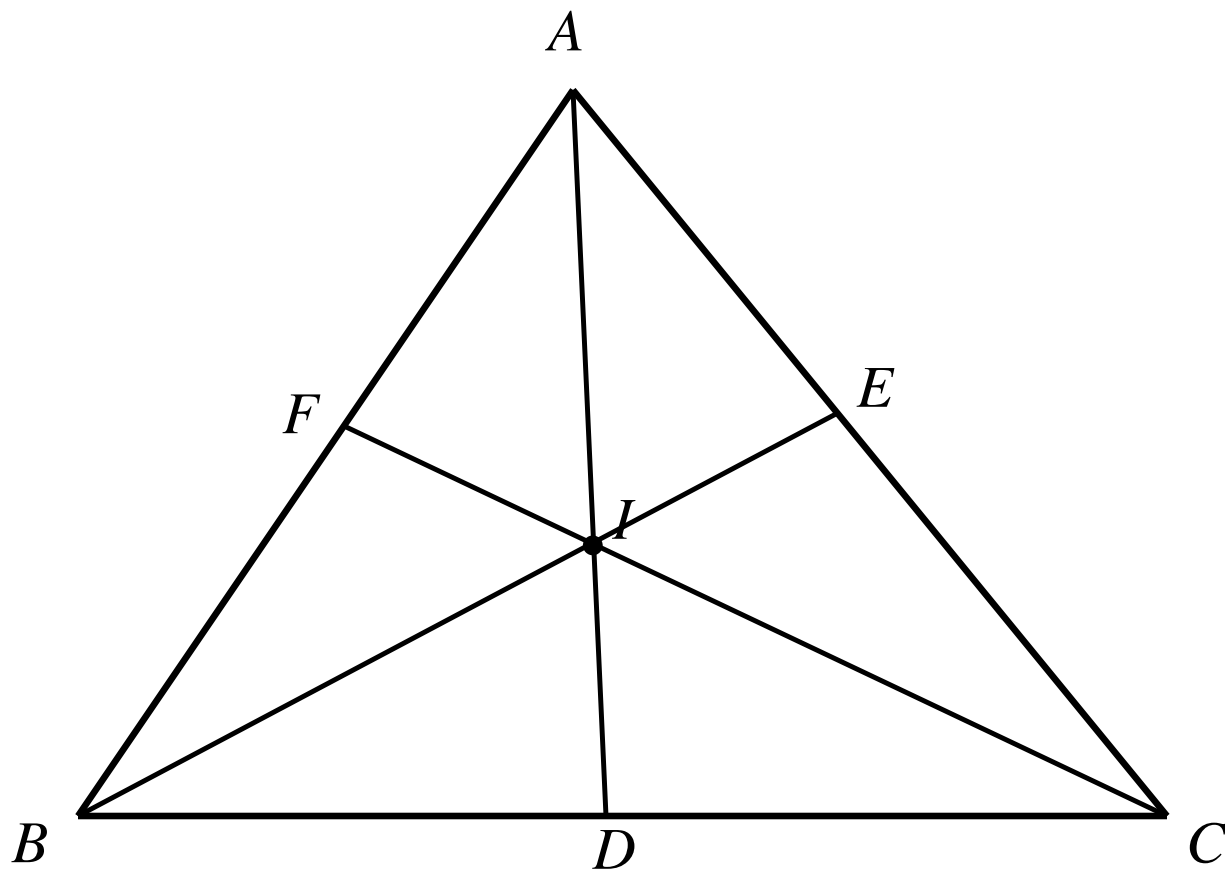
Proof:

- Since D is the midpoint of BC , we have $\frac{BD}{DC} = 1$. Similarly, we have $\frac{CE}{EA} = 1$ and $\frac{AF}{FB} = 1$. Thus, the Ceva product is

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1$$

- By the converse of Seva's theorem (Theorem 4), the three medians are concurrent. ■

Theorem 6: The three bisectors of a triangle intersect at a single point (the **incenter**).



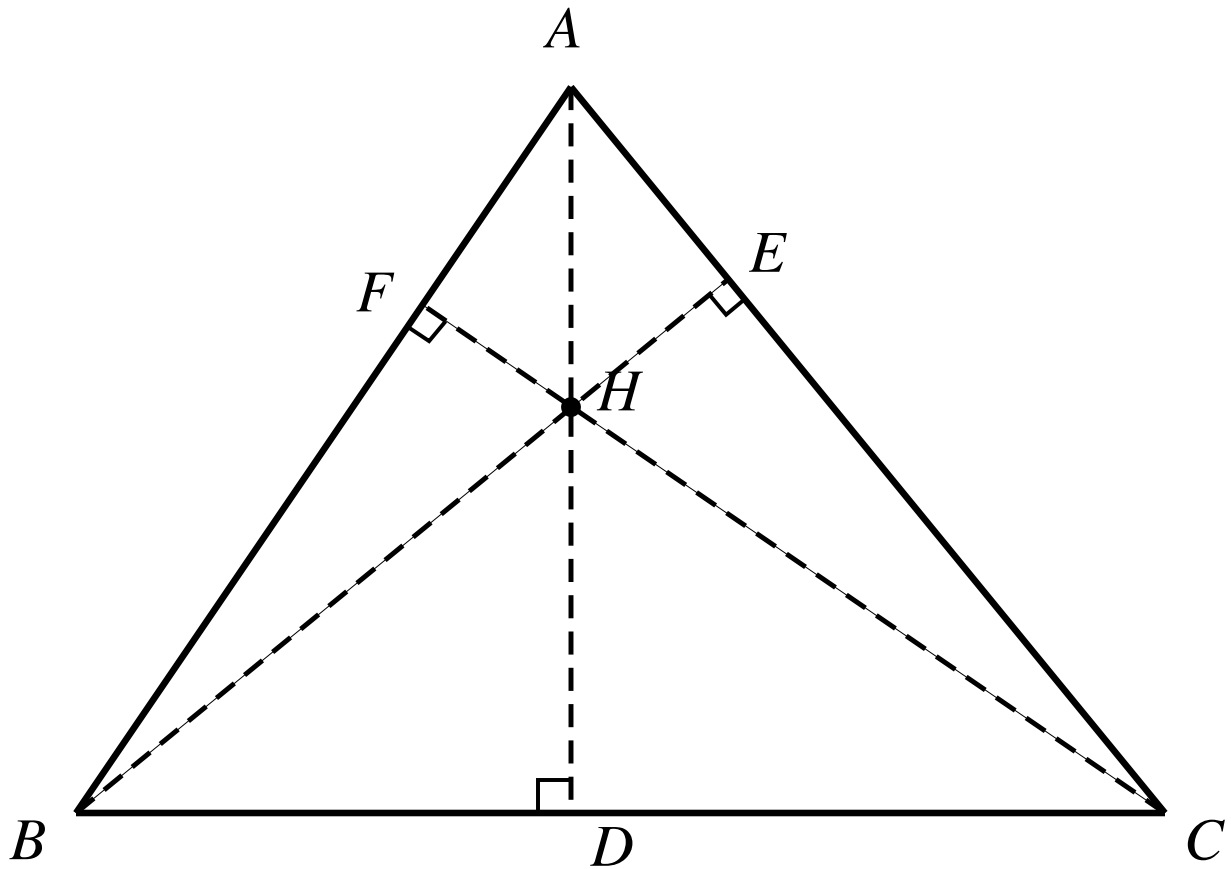
Proof:

- The bisector theorem gives us $\frac{BD}{DC} = \frac{AB}{AC}$, $\frac{CE}{EA} = \frac{BC}{AB}$ and $\frac{AF}{FB} = \frac{AC}{BC}$. Multiplying the three equations together, we get

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1$$

- By the converse of Seva's theorem (Theorem 4), the three bisectors are concurrent. ■

Theorem 7: The three altitudes of a triangle intersect at a single point (the **orthocenter**).



Proof:

- Let D, E, F be the feet of the altitudes from A, B, C to sides BC, CA, AB . From the right triangles at each foot: $BD = c \cos B$, $DC = b \cos C$, $CE = a \cos C$, $EA = c \cos A$, $AF = b \cos A$, $FB = a \cos B$.
- Thus the Ceva product is

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = \frac{c \cos B}{b \cos C} \cdot \frac{a \cos C}{c \cos A} \cdot \frac{b \cos A}{a \cos B} = 1.$$

- By the converse of Ceva's theorem (Theorem 4), the three altitudes are concurrent. ■

Remark:

- the orthocenter can be outside the triangle, but the altitudes still intersect at a single point. So Ceva's theorem does not directly imply the concurrency of altitudes.
- Formula the Ceva's theorem for a point outside the triangle, and use it to show that the altitudes of an obtuse triangle are concurrent.